

Lab 1 Submission

Docker Environment Screenshots

1. OpenCL and device query output.

```

ubuntu@cd0589796b82:/workspaces/wes237b/assignment-1/lab/device_query$ clinfo -l
clinfo: /lib/aarch64-linux-gnu/libOpenCL.so.1: no version information available (required by clinfo)
Platform #0: QUALCOMM Snapdragon(TM)
  -- Device #0: QUALCOMM Adreno(TM) 643
Platform #1: Portable Computing Language
  -- Device #0: cpu-neoverse-n2-cortex-a55
ubuntu@cd0589796b82:/workspaces/wes237b/assignment-1/lab/device_query$ make run
Finding OpenCL Platforms...
Found 2 platforms!
Platform 0
- Name: QUALCOMM Snapdragon(TM)
- Version: OpenCL 3.0 QUALCOMM build: 0838.2
- Vendor: QUALCOMM
- Num Devices: 1
  Device: 0
  - Name: QUALCOMM Adreno(TM) 643
  - Type: GPU
  - Compute units: 2
  - Global memory size: 2775556096 bytes (2646 MB)
  - Max constant buffer size: 65536 bytes (0 MB)
  - Max local memory size per Compute Unit: 32768 bytes (32 KB)
  - Max Work Item Dimensions: 3
  - Max Work Item size: 1024 x 1024 x 1024
  - Max group size: 1024

Platform 1
- Name: Portable Computing Language
- Version: OpenCL 3.0 PoCL 7.1 Linux, Release, RELOC, LLVM 20.1.8, SLEEF, POCL_DEBUG
- Vendor: The pocl project
- Num Devices: 1
  Device: 0
  - Name: cpu-neoverse-n2-cortex-a55
  - Type: CPU
  - Compute units: 8
  - Global memory size: 4163334144 bytes (3970 MB)
  - Max constant buffer size: 524288 bytes (0 MB)
  - Max local memory size per Compute Unit: 524288 bytes (512 KB)
  - Max Work Item Dimensions: 3
  - Max Work Item size: 4096 x 4096 x 4096
  - Max group size: 4096

```

2. Raytracer sequential output.

```

ubuntu@cd0589796b82:/workspaces/wes237b/assignment-1/lab/raytracer_sequential$ make run
gcc -O2 -std=c99 -o raytracer_sequential main.c lib/geometry/Sphere.c lib/vec_ops.c lib/geometry/Light.c -lm
./raytracer_sequential
Starting Sequential Ray Tracing...
=====
Single-Threaded Implementation on CPU
Total execution time (host + kernel): 2259.226 ms
=====

```

3. Raytracer parallel output

```

make: *** [Makefile:19: gpu] Error 1
ubuntu@cd0589796b82:/workspaces/wes237b/assignment-1/lab/raytracer_parallel$ make
make: Nothing to be done for 'all'.
ubuntu@cd0589796b82:/workspaces/wes237b/assignment-1/lab/raytracer_parallel$ ls
Makefile core kernel.cl lib main.c raytracer_parallel
ubuntu@cd0589796b82:/workspaces/wes237b/assignment-1/lab/raytracer_parallel$ ./raytracer_parallel 0
=====
Device: QUALCOMM Adreno(TM) 643
Total execution time (host + kernel): 304.507 ms
Image titled output_gpu.png has been created/modified and can now be viewed!
=====

```